A068 v0.4.4 — PKLSA Intrinsic Phase Clock Material Tubes, Full-field RPO and Floquet Falsifier

SST-Workbench | Framework v1.0.10

19 October 2026 – development / physical inference unresolved

Abstract

This version *actually evolves* a fully three-dimensional unforced incompressible Euler velocity field initialized from a SHA-256-identified native KnotPlot trefoil, using the unchanged A064 Euler solver, finite-core regularization, Fourier truncation and RK4. Reconstructed 3D vorticity yields an enstrophy-weighted shape coordinate and a separate enstrophy/energy functional; neither has an imposed frequency. A new Lagrangian torus is advected through the same RK4 stages as the actual Euler field, with local vorticity-tangency and material-volume diagnostics. Complete 3D velocity fields are searched for returns under three periodic translations and the 24 proper cubic rotations. The exact A064 tangent operator is separately finite-difference verified; Floquet Ritz prediction is strictly conditional on RPO and material-tube screening. Space/time/core refinement and framework provenance are retained. A separately preregistered 2015 notebook A/B geometric seed atlas is additionally audited and used for controlled short Euler pilot evolutions, without promoting source shape differences to physical-provider independence. This is a **numerical model** initialized by PKLSA/KnotPlot geometry; it is *not* a confirmed source-native material vortex or an independent physical PKLSA clock. Previously inspected geometry and past diagnostics remain **unblinded development evidence**; official BLIND packaging does not retroactively restore epistemic blindness.

1 Identification and certification boundaries

Field	Value
Catalog	A068
Version	v0.4.4
Framework	SST Falsifier Framework v1.0.10 candidate
Profile	`multilibrary_gpu`
Current authority	`EULER_MODEL_SIMULATION_ONLY`
Physical clock	`UNRESOLVED`

2 Research question and objective

Can an autonomous, unforced 3D Euler vortex initialized from native trefoil geometry generate a reproducible intrinsic phase without inserting a clock frequency? What additional independent tests are needed to claim a *material, physical PKLSA* clock? The present version is a numerical prerequisite, not a physical verification.

3 Competing hypotheses and decision semantics

H_0: a numerical transient, slow vortex deformation, rigid-body phase, inadequate sampling, nonconvergent core/mesh effects, or operator-fitting artefact mimics a clock. H_1: a genuine source-native material vortex has a stable relative periodic orbit, coherent intrinsic oscillatory phase and independent withheld operator prediction. An unqualified source produces `UNRESOLVED`, not physical `FAIL`; a failed numerical discretization is not an SST-theory falsification.

4 Assumptions and limiting cases

The 47-point `knot_3.1.txt` geometry belongs to one KnotPlot-native family. Its numerical centerline has no authenticated Euler-equilibrium relaxation, no independently verified source-native topology and no SI radius. The original A064 centerline is placed in a periodic cubic domain of length $L = 2\pi$ after arbitrary normalization; Gaussian smoothing of its line vorticity uses $a = 0.4$ nominal units and $\Gamma = 1$ model unit. The finite-core filter is a numerical modeling assumption, *not* a derivation of canonical r_c. Actual 3D velocity and vorticity evolve without a driving frequency. A Hilbert-transform phase of a transient need not be a stable clock. A transported surface is material by construction but is a vortex tube only if $\boldsymbol{\omega}\cdot\mathbf{n}$ is suitably small on the surface. A low full-field relative return is a candidate, not a Newton–Krylov shooting certification.

5 Symbols, dimensions and units

Symbol	Meaning	SI dimensions if calibrated
$\mathbf{u}, \boldsymbol{\omega}$	velocity, curl of velocity	LT^{-1}, T^{-1}
L, a, Γ	box length, core width, circulation	L, L, L^2T^{-1}
E	mean energy per unit mass	L^2T^{-2}
H	mean helicity	LT^{-2}
Z	mean enstrophy	T^{-2}
C, λ_i	covariance and its eigenvalues	L^2
q_{sh}, q_Z	two dimensionless functionals	1
$\phi, \widehat{\Omega}$	observed phase, angular slope	$1, T^{-1}$
$\mathbf{X}, \mathbf{n}$	material nodes, unit normal	$L, 1$
T_R, g	relative return period, discrete/continuous symmetry	$T, 1$
M_g, μ	relative monodromy, multiplier	$1, 1$

No dimensional conversion to s, m or kg is made. In particular no SST circulation scale or relativistic time-dilation law is used to set a frequency.

6 Actual continuous and numerical model

The finite-dimensional Galerkin operator implements the projected Lamb form of Euler:

$$\partial_t \mathbf{u} = \mathcal{P}[\mathbf{u}\times\boldsymbol{\omega}], \quad \boldsymbol{\omega} = \nabla\times\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0. \tag{1}$$

The original A064 code deposits a signed line-current along each polygon edge (cloud-in-cell), applies Gaussian finite-core smoothing and a two-thirds wavenumber cutoff, then constructs the divergence-free Fourier Biot–Savart velocity

$$\widehat{\mathbf{u}}_0(\mathbf{k}) = \frac{i\mathbf{k}\times\widehat{\boldsymbol{\omega}}_a(\mathbf{k})}{|\mathbf{k}|^2} \quad (\mathbf{k}\neq 0). \tag{2}$$

RK4 evolves this field without inserted phase. Conservation screens use

$$E = \frac{1}{2V}\int_V |\mathbf{u}|^2\,dV, \quad H = \frac{1}{V}\int_V \mathbf{u}\cdot\boldsymbol{\omega}\,dV, \quad Z = \frac{1}{2V}\int_V |\boldsymbol{\omega}|^2\,dV. \tag{3}$$

For the full periodic volume, use enstrophy weights to construct the periodic-centered three-dimensional covariance tensor C. Ordered eigenvalues define the shape scalar

$$q_{\mathrm{sh}} = \frac{\lambda_1-\lambda_2}{\operatorname{tr} C}, \quad q_{\mathrm{sh},2} = \frac{\lambda_2-\lambda_3}{\operatorname{tr} C}. \tag{4}$$

These two eigenvalue gaps are correlated. Instead of treating them as independent physical clocks, use the distinct flow functional

$$q_Z = \frac{L^2 Z}{E}. \tag{5}$$

Both q_{sh} and q_Z are computed from each actual evolved 3D field, not from imposed oscillators. The analytic-signal phase estimator is

$$\phi_q(t) = \operatorname{unwrap} \arg\{[q(t) - \bar{q}] + i\mathcal{H}[q(t) - \bar{q}]\}, \qquad \widehat{\Omega}_q = \operatorname{slope}_W \phi_q(t). \tag{6}$$

This estimator is a screen only; it does not establish a relative periodic orbit or intrinsic clock. Thresholds were specified in method preflight: at least six unwrapped cycles; RMS phase-fit error at most 0.20 radians; instantaneous frequency CV at most 0.12; cross-frequency relative error at most 0.03. A constant shape under rigid rotation is no clock. A few cycles of a windowed transient are not a proof of stability. Time series are uniformly sampled by construction.

v0.4.4 material and stability operators

The finite-core tube seeded around the native trefoil is a *candidate* material surface. At every Euler RK4 substage its nodes obey

$$\frac{d\mathbf{X}}{dt} = \mathbf{u}(\mathbf{X}, t), \quad \eta_\omega = \frac{1}{A}\int_S \frac{|\boldsymbol{\omega}\cdot\mathbf{n}|}{\max(|\boldsymbol{\omega}|, \epsilon_\omega)}\, dA, \quad \delta_V = \frac{|V(t) - V(0)|}{|V(0)|}. \tag{7}$$

The prescribed screening limits are $\delta_V < 0.05$, area-mean $\eta_\omega < 0.05$, and pointwise maximum $|\widehat{\omega}\cdot\mathbf{n}| < 0.10$, at all retained snapshots. Net vorticity flux through a closed surface is *not* an admissible vortex-tube test; it vanishes even on surfaces not made of vortex lines. Initial tube parameters are fixed at 32 axial sections, 10 azimuthal sides and radius 0.12 model units.

The full-field symmetry-reduced return statistic is

$$r_{\mathrm{RPO}}(T, g) = \frac{\|\mathbf{u}(T) - g\mathbf{u}(0)\|_{L^2(V)}}{\|\mathbf{u}(0)\|_{L^2(V)}}, \qquad g \in \mathbb{T}^3 \rtimes O^+_{\mathrm{cube}}. \tag{8}$$

The implementation registers a 0.025 relative candidate screen and searches all 24 proper cubic rotations with 3D periodic translations. This is a *full 3D-field search*, not proof of a converged nonlinear RPO. Repeated returns, neutral drift modes, shooting and mesh/time convergence remain necessary.

The independent tangent equation is evaluated using the unmodified A064 Euler linearization,

$$\dot{\delta\mathbf{u}} = \mathcal{P}[\delta\mathbf{u}\times\boldsymbol{\omega} + \mathbf{u}\times(\nabla\times\delta\mathbf{u})], \qquad M_g = g^{-1}D\Phi^{T_R}(\mathbf{u}_0), \tag{9}$$

where $M_g v = \mu v$, $\Omega_F = |\arg\mu|/T_R$ and $\epsilon_F = \|M_g v - \mu v\|/\|M_g v\|$. This is an *operator* frequency; a measured q-phase never enters the tangent propagator. Ritz values from a five-vector fixed-seed subspace are diagnostics only. No physical Floquet frequency can be reported without a validated RPO, residual, spectral convergence and a withheld prediction comparison.

7 Executable algorithm and fixed run matrix

1. Verify byte-exact historical controls and A064 solver/source hashes; refuse changed or missing sources.

2. Load 47 native 3D XYZ vertices; normalize and map to the periodic box; deposit vorticity and compute solenoidal $\mathbf{u}(0)$ from A064.

3. In `BASIC`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 6$, retaining 151 genuine 3D field snapshots and advected material mesh states.

4. In `FULL/CERTIFY`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 20$, retaining 251 3D fields and synchronously advected material surfaces. Independently evolve $N = 24$ at identical $(a, \Delta t, T)$, $N = 16$ at $\Delta t = 0.02$, and $N = 16$ with $a = 0.32, 0.48$ at fixed $(T, \Delta t)$.

5. At every retained field, measure material-volume drift, surface vorticity tangency, and E, H, Z, divergence, CFL, shape tensor, q_{sh} and q_Z. Recover phase only from measured functionals; do not input a theoretical frequency.

6. Evaluate complete Euler state returns up to the 24 cubic rotations and 3D translations, retaining top RPO candidates and relative L^2 residues. Validate tangent-linear Euler against finite differences and a manufactured analytic two-mode Floquet oracle. Only attempt matrix-free Ritz on admissible candidates.

7. Require numerical conservation to close G2; record spatial/time/core diagnostic differences even when downstream physical gates are blocked. Do not promote G3 unless material-tube tangency, a verified nonlinear relative periodic orbit, independent providers and independently predicted frequency all qualify.

8. Let the official v1.0.10 runtime write unique BLIND ZIPs, PDFs and gate-ledger hashes; do not bypass reveal policy.

8 Source provenance and provider independence

The 40 historical XYZ curves are imported exclusively from the externally installed, SHA-256-pinned `A004_historical_trefoil/SST_Historical_Trefoil_XYZ_v0.2.2/data` release under `02_libraries/A_knot_libraries`. There are no internal A068 copies or automatic fallback; missing or changed A004 data fails G1 closed. The orientation reversal holds the first vertex fixed. Historical A/B shapes and controls do not provide independent physical PKLSA dynamical evidence.

`knot_3.1.txt`: SHA-256 `98a628af4ba5d09274f0a1845c351a7d633a39ff833a3ae0b78e351bafdabf`. The original A064 Python Euler solver is included byte-for-byte at `experiment/euler3d.py` and as historical reference. Earlier two A068 preflight JSONs and A064-v0.3.0 results are immutable controls, not independent vortex providers. The finite-core smoothing function, Fourier truncation and normalization are inherited from A064. The expected upstream hashes are enforced in G1, and mesh/core variation does not create distinct provider families. Input XYZ geometry has *unverified* physical vortex equilibrium and no SI units. A tracer torus, RPO scan, or tangent derivative constitutes a distinct numerical operator, not a distinct physical provider.

v0.4.4: Full diagnostic program and GPU authority

This instance uses Framework v1.0.10 as a **LOCAL_VALIDATED_CANDIDATE**, not canonical hardware certification. Independently executable scientific gates use FULL_DIAGNOSTIC scheduling: an earlier numerical FAIL cannot erase later measured diagnostic results, but downstream claims remain DIAGNOSTIC_ONLY when prerequisites are unqualified. No REVEAL is permitted without an independently qualified final gate.

Tranche M: The same Euler RK4 stage states advect three material meshes (32×10, 64×20, 128×40). A separate set of nearby tracers estimates $F_{ij} = \partial X_i / \partial X_{0j}$ and $\det F$. Mesh degeneracy is reported, while self-intersection/embeddedness remains unverified.

Tranche V: Candidate vortex surfaces are generated by integrating $dX/ds = \omega/|\omega|$ from transverse seed rings. End-to-start closure and local $\omega \cdot n$ are measured: failed closure is not repaired by forcing a seam.

Tranche R: Full-field RPO candidates may enter fixed-group fixed-period Newton–GMRES shooting, a second independently reintegrated return and symmetry-residual measurements. Optimizing the free period and group phase, neutral-mode gauge fixing and spatial convergence remain independent certification requirements.

Tranche F: The Euler tangent propagator and matrix-free Arnoldi eigenpairs are independent of the measured scalar phase. Real Floquet frequencies are strictly conditional on an admissible RPO; the manufactured two-mode eigenvalue is a software test only.

Tranche GPU: A genuine three-dimensional FFT/RK4 solver using PyTorch XPU on Intel Arc runs as a separate `float32` screening lane and conditional `float64` parity lane. Every run must record actual XPU device, full-state L2 error against independently integrated NumPy FP64, energy and helicity differences. Missing XPU gives `UNAVAILABLE_NO_FALLBACK`, never a CPU substitute. Unsupported FP64 is separately recorded while FP32 remains screening only. Framework SYCL FP32/DD32 common-kernel parity is *separate* and must be hardware-tested on the target Arc A770 before v1.0.10 canonical promotion.

Registered cost bound: BASIC mesh deformation $T = 0.24$, FULL $T = 0.80$, CERTIFY $T = 0.80$ with halved step; main A068 Euler campaigns retain their longer trajectories. These mesh studies are numerical prescreens, not full physical time convergence. All 40 external A004 v0.2.2 XYZ hashes and older A064 files remain unchanged.

Additional registered operators (E16–E22).

$$\dot{F} = (\nabla \mathbf{u})F, \qquad J = \det F, \tag{10}$$

$$\delta_{V,h} = |\delta V_{h/2} - \delta V_h|, \tag{11}$$

$$\frac{d\mathbf{X}}{ds} = \frac{\boldsymbol{\omega}(\mathbf{X})}{|\boldsymbol{\omega}(\mathbf{X})|}, \tag{12}$$

$$\mathbf{R}(U_0, T, g) = g^{-1}\Phi_T(U_0) - U_0, \tag{13}$$

$$r_2 = \frac{\|g^{-2}\Phi_{2T}(U_0) - U_0\|}{\|U_0\|}, \tag{14}$$

$$M_g\mathbf{v} = \mu\mathbf{v}, \qquad \epsilon_F = \frac{\|M_g\mathbf{v} - \mu\mathbf{v}\|}{\|M_g\mathbf{v}\|}, \tag{15}$$

$$\delta_{\mathrm{GPU}} = \frac{\|\mathbf{u}_{\mathrm{XPU}}(T) - \mathbf{u}_{\mathrm{FP64}}(T)\|_2}{\|\mathbf{u}_{\mathrm{FP64}}(T)\|_2}. \tag{16}$$

All ratios are dimensionless. F is an operational deformation-gradient estimate, not an independently reconstructed physical vortex equilibrium. M_g is conditional on a qualified relative return; no observed Hilbert frequency enters the operator.

Historical A/B trefoil geometric seed atlas (new v0.4.4)

Two distinct 2015 notebook-parametric trefoils are provided as headerless, byte-preserved XYZ arrays, not as physical PKLSA vortex equilibria:

$$\mathbf{X}_A(t) = \big(\cos t + 1.5\cos 2t,\ \sin t - 1.5\sin 2t,\ 0.35\sin 3t\big), \tag{17}$$

$$\mathbf{X}_B(t) = \big((2+\cos 3t)\cos 2t,\ (2+\cos 3t)\sin 2t,\ \sin 3t\big), \quad t \in [0, 2\pi). \tag{18}$$

Analytic reference lengths are $\ell_A = 19.976521990340952$ and $\ell_B = 31.898600666412335$ historical coordinate units. Each has $N \in \{600, 1200, 2400, 4800, 9600\}$ with original/reversed and X-mirrored/reversed controls: $2 \times 5 \times 4 = 40$ immutable files. Each vertex file has implicit periodic closure; the true equal-arc-length sampling and SHA-256 are checked against the A004 v0.2.2 manifest. Reversal of polygon traversal reverses the imposed line-current direction in the inherited Euler initializer; $x \mapsto -x$ changes chirality but is not an independent physical realization.

$$\ell_N = \sum_{j=0}^{N-1} \|\mathbf{X}_{j+1} - \mathbf{X}_j\|, \quad CV_e = \frac{\mathrm{sd}(\ell_j)}{\mathrm{mean}(\ell_j)}, \qquad \mathbf{X}_N = \mathbf{X}_0. \tag{19}$$

At fixed Galerkin grid size n, core a, Δt and T, $\delta_{Z,\mathrm{seed}} = |Z_{N600}(T) - Z_{N1200}(T)| / \max(|Z_{N1200}(T)|, \epsilon_Z)$ measures *source-centerline sampling* error. It is explicitly not Euler grid convergence. BASIC runs

two short A/B geometry-initiated 3D Euler pilots; FULL/CERTIFY separately run eight preregistered A/B sampling, mirror and traversal controls. For the A/B original N600 candidates in FULL/CERTIFY, an advected toroidal surface, full-vector endpoint translation screen (identity rotation only) and separately finite-difference-verified Euler tangent operator are also computed. This historical sidecar does *not* perform 24-rotation RPO shooting or a certified Floquet prediction; those are distinct primary A068 operators and require qualified physical inputs. All original XYZs are dimensionless, non-SI and geometrical only. **Pinned A004 source:** the preceding v0.4.1 development archive reported an outer-v0.2.0 / inner-v0.1.0 provenance inconsistency. The present v0.4.4 release uses instead the separately verified A004 `v0.2.2` manifest (SHA-256 `e7477a9c817f59124f279d08f001f5ef7f85c0a7c178e0ecf5b0b0116e57083e`) and all 40 external XYZ hashes. Its `cw` ordering preserves the first vertex. A/B geometry and controls contribute **zero independently qualified physical PKLSA providers** and cannot independently close G3–G8.

9 Dependency graph and physical guardrails

Gate	Required decision	Initial status
G0	Framework freeze, signed provenance path	runtime
G1	Native geometry, solver and historical control hashes	source check
G2	Real Euler 3D evolution plus $E, H, \nabla \cdot \mathbf{u}$, CFL	numerical PASS/FAIL
G3	Six clean cycles; *material tube and RPO independently qualified*	UNRESOLVED
G4	Independent withheld Euler/Floquet prediction	blocked
G5	Converged space, time and core with physical state fixed	blocked
G6	Explicit native backend parity/authority	blocked
G7	Multiple independent qualified PKLSA providers	blocked
G8	Physical mechanism conclusion if all required gates pass	blocked
G9	Nonced reveal commitments on allowed frozen gate ledger	withheld

Gate order is an explicit DAG. A negative numerical screen does not count as a physical hypothesis falsification. A gapped clock phase without an identified finite-core material vortex can *never* close G3 in this release.

10 Numerical control and exploratory observations

For G2 preregistered development thresholds: relative divergence $< 10^{-9}$; relative energy drift $< 10^{-5}$; relative helicity drift $< 10^{-4}$; CFL < 0.25; finite velocity field, and nontrivial evolution. Comparative tests use

$$\delta_Z(A, B) = \frac{|Z_A(T) - Z_B(T)|}{\max(|Z_B(T)|, 10^{-30})}. \tag{20}$$

Exploratory, **unblinded** local FULL prechecks (not an official frozen framework run):

Case	Grid, core, Δt	$Z(T)$	$\lvert\Delta E/E\rvert_{\max}$	Phase cycles
Base	$16, 0.40, 0.04$	0.064202	8.34×10^{-10}	0.646
Spatial	$24, 0.40, 0.04$	0.121697	6.93×10^{-9}	0.615
Temporal	$16, 0.40, 0.02$	0.064202	4.22×10^{-11}	0.646
Core low	$16, 0.32, 0.04$	0.094750	1.69×10^{-9}	0.643
Core high	$16, 0.48, 0.04$	0.035642	2.04×10^{-10}	0.723

δ_Z under time-step halving is 2.01×10^{-9}, while spatial $\delta_Z \simeq 0.472$ and core-spread $\simeq 0.921$. Thus temporal integration control succeeds but spatial/core physical convergence demonstrably **does not** qualify. Less than one observed phase cycle fails the method screen's six-cycle minimum; it cannot be

extrapolated to a period by assumption. These values are preliminary numerical model diagnostics and will be recomputed on each framework run.

11 Backend authority

The primary reference field integration is the upstream NumPy/SciPy FP64 Euler simulator. An optional independent torch-XPU full 3D Euler lane reports actual device and field parity; its presence does not certify a physical vortex. Its runs are not equivalent to native C++ FP64 or Intel SYCL parity certification. Framework v1.0.10 backend configuration may disable certain gates; disabled scientific gates do not pass. DD32/FP32 only screen, never physical confirmation. All core/mesh branches here share one upstream mathematical method and source provenance.

12 Blindness, frozen protocol and reveal

The historical source and this exploratory run have already been inspected, so only framework *output confidentiality and audit integrity* are blind. Future provider-independent PKLSA-clock claims must be preregistered on a fresh held-out source before opening it. Frozen v1.0.10 publishes run-specific ZIP/PDF identities and archives previous output roots; `REVEAL_IF_ALLOWED` correctly refuses when a required physical G8 is diagnostically `UNRESOLVED`. Never edit a frozen policy, ledger or HMAC commitment to force a reveal.

13 Statistics and confidence limits

The Euler trajectory is one deterministic single-provider numerical realization. Time snapshots, mesh steps and smoothing variants are *not* independent experimental replicates. A Hilbert phase slope on less than one cycle has no established asymptotic period. No physical confidence interval or p-value is computed. Discovery windows and candidate selection are fixed by code; no frequency target, SST phase law or operator eigenvalue is supplied.

14 Actual runtime results and retained evidence

Each official BASIC/FULL/CERTIFY pipeline computes and stores `EULER_DYNAMICS_RESULTS.json`, `EULER_3D_FIELDS_MODEL_ONLY.npz`, `MATERIAL_MESH_CONVERGENCE_AND_CAUCHY.json`, `LONG_HORIZON_MATERIAL_TUBE_V044.json`, `VORTEX_LINE_RETURN_MAP_V044.json`, `GPU_XPU_ENVIRONMENT_PREFLIGHT_V044.json`, `VORTEX_LINE_SEED_SCREEN.json`, `NONLINEAR_RPO_SHOOTING_AND_SECOND_RETURN.json`, `INDEPENDENT_FLOQUET_ARNOLDI.json`, and `GPU_XPU_FULL_EULER_RK4_PARITY.json`. The `HISTORICAL_2015_EULER_MODEL_DIAGNOSTICS.json` remains a geometric-seed sidecar. The official framework writes hashed source/run manifests, a gate ledger, PDF and BLIND ZIP for each run. A BASIC result cannot substitute for FULL Euler mesh/core convergence. No GPU or physical-clock conclusion follows from an absent XPU device or an operator oracle alone.

Runtime gate ledger will be inserted by frozen framework execution.

15 Interpretation and next falsification step

The existence of nonlinear Euler evolution does not demonstrate a stable toroidal material boundary, spectral convergence, global RPO or an intrinsic physical clock. Future v0.4 qualification needs a source-native independent core field, advected material boundary with demonstrated low flux, full-state recurrence modulo allowed symmetries, a separately computed time-dependent Floquet/Koopman prediction and a new source/protocol commitment. Historical A064 spectral eigenpairs at $N = 8, 12, 16$ were nonconvergent and cannot be used as the predicted clock frequency here. Likewise, no comparison to Lorentz time dilation is permitted before a physical intrinsic oscillator is certified.

16 Reproducibility and bibliography

Install only into a *new* A068-v0.4.4 instance with the official `tools/new_falsifier.py` from locally present v1.0.10. Run `pytest -q tests`, then `run_all.cmd FREEZE`, `run_all.cmd BASIC`, `run_all.cmd FULL`, and optionally `run_all.cmd CERTIFY`; no frozen package from v0.2.0 is overwritten. Preserve all v0.2.0 result evidence and commits.

References

[1] P. G. Saffman, *Vortex Dynamics* (Cambridge University Press, 1992), doi:10.1017/CBO9780511624063.

[2] H. K. Moffatt, J. Fluid Mech. **35**, 117–129 (1969), doi:10.1017/S0022112069000991.

[3] D. Gabor, J. Inst. Elect. Eng. Part III **93**, 429–457 (1946), doi:10.1049/ji-3-2.1946.0074.

17 v0.4.4 registered long-horizon diagnostic addendum

The FULL profile advances all three registered toroidal material meshes using identical Euler–RK4 stages through $T = 20$ model time. The independently computed volume histories $\delta_V^{(h)}$ and satellite $\det F$ are screened at a finite set of snapshots. A sampled triangle contact broadphase may identify potential mesh intersections but cannot prove embeddedness. The registered multi-anchor vorticity return map reports genuine integral curve endpoint errors before any closure. GPU measurements use the actual XPU device, if available, and never substitute a CPU lane. None of these implementation-level improvements qualifies a source-native PKLSA clock.